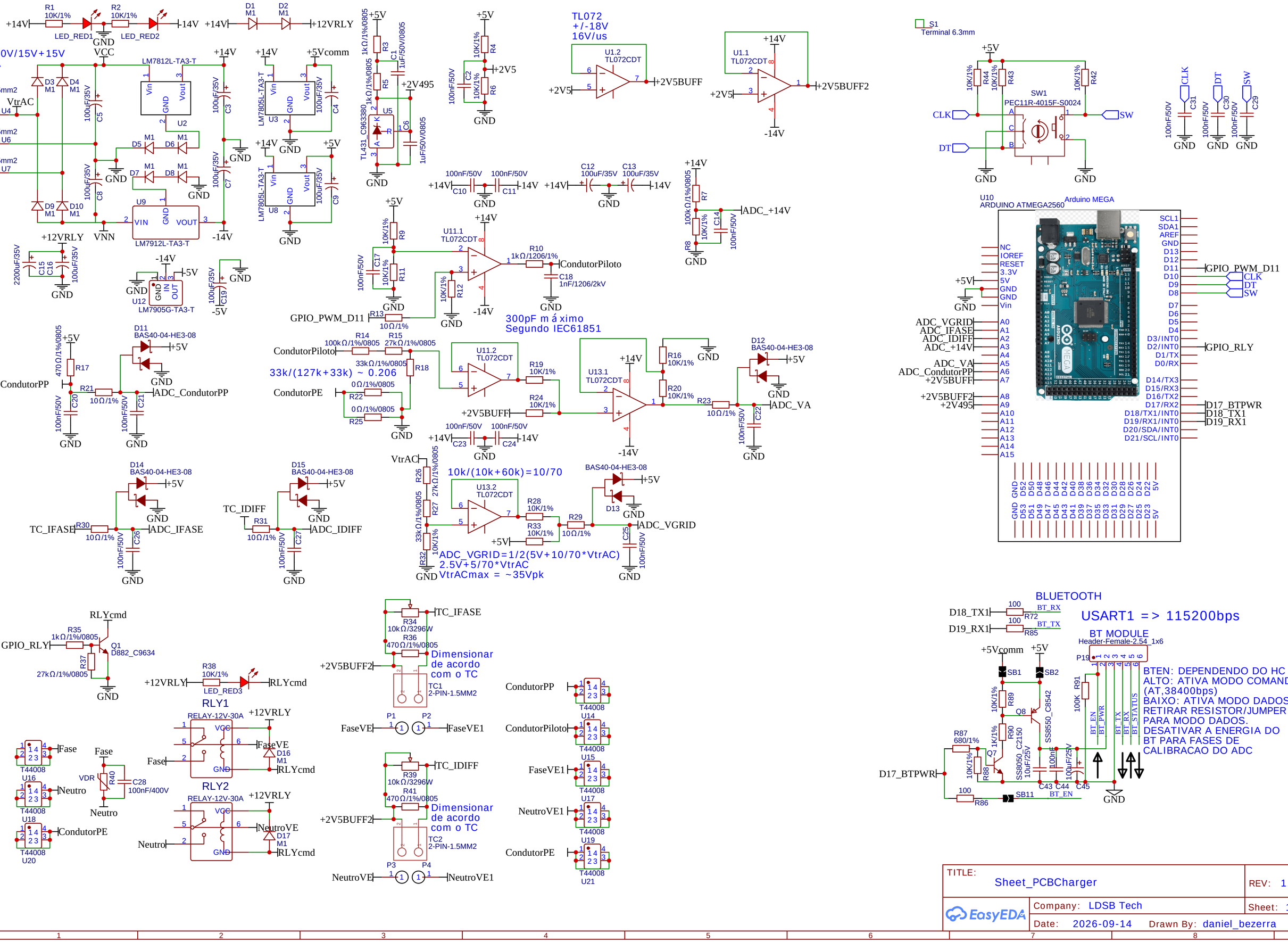
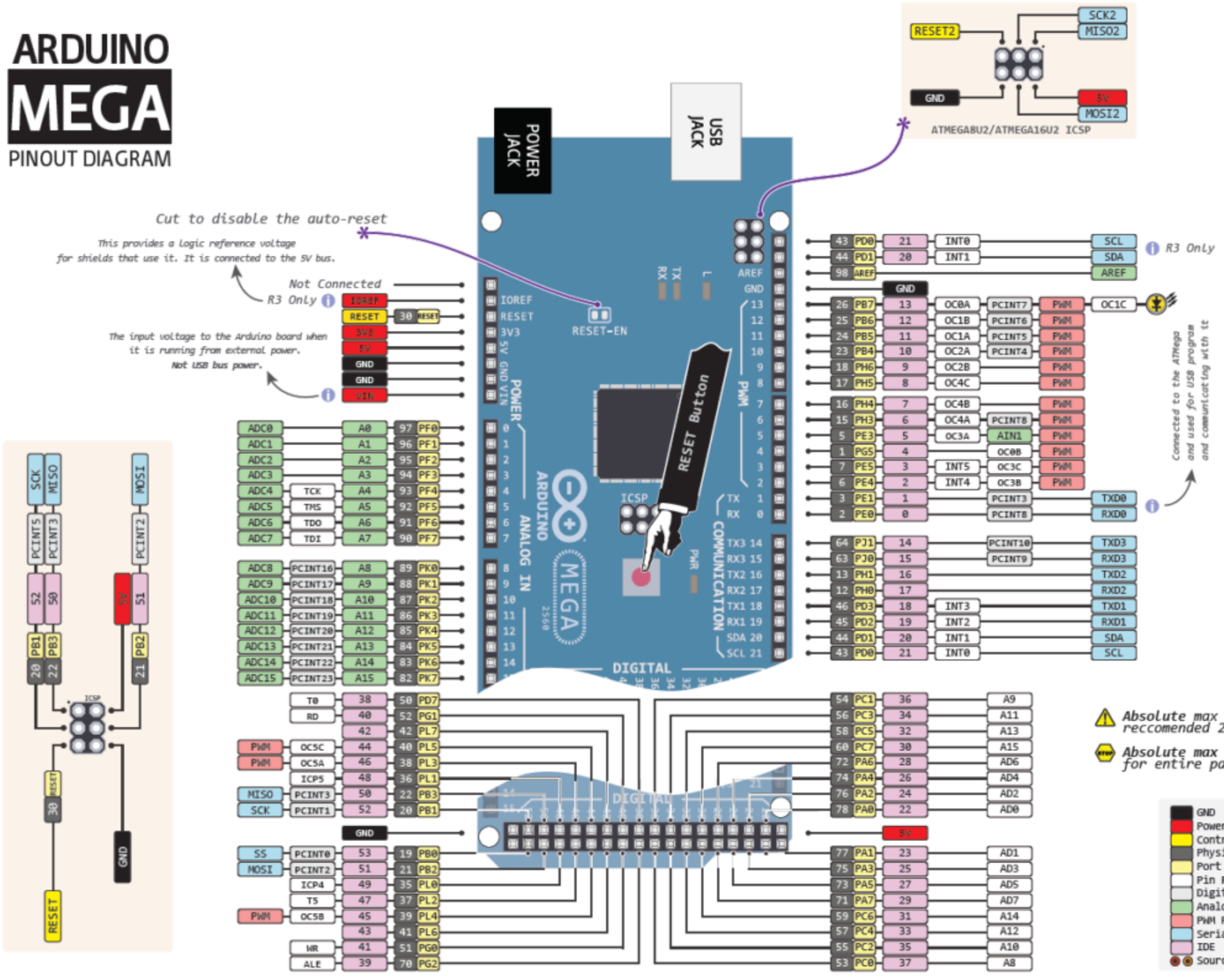

R1 10K/1%
R2 10K/1%
LED_RED1
LED_RED2
+14V
-14V
GND
D1 M1
D2 M1
+12VRLY
220V/15V+15V
1A
VCC
LM7812L-TA3-T
VtrAC
T_1.5mm2
1U4
1U6
1U7
D3 M1
D4 M1
D5 M1
D6 M1
D7 M1
D8 M1
D9 M1
D10 M1
100uF/35V
C3
C4
C5
C7
C8
U2
U3
U8
U9
LM7805L-TA3-T
LM7912L-TA3-T
VIN
VOUT
VNN
+5Vcomm
2200uF/35V
C15
C16
C19
U12
LM7905G-TA3-T
IN
OUT
-5V
+5V
1kΩ/1%/0805
R3
R5
C1
1uF/50V/0805
+2V495
U5
C963380
TL431
C6
1uF/50V/0805
10K/1%
R4
R6
C2
100nF/50V
+2V5
TL072
+/-18V
16V/us
U1.2
TL072CDT
+2V5BUFF
U1.1
+2V5BUFF2
C10
C11
C12
C13
100uF/35V
R7
R8
C14
100nF/50V
100kΩ/1%/0805
ADC_+14V
R9
R11
C17
R10
1kΩ/1206/1%
CondutorPiloto
C18
1nF/1206/2kV
U11.1
U11.2
R12
R13
10Ω/1%
GPIO_PWM_D11
300pF máximo
Segundo IEC61851
R14
100kΩ/1%/0805
R15
27kΩ/1%/0805
33kΩ/1%/0805
R18
CondutorPE
0Ω/1%/0805
R22
R25
33k/(127k+33k) ~ 0.206
R19
10K/1%
R24
10K/1%
U13.1
R16
10K/1%
R20
10K/1%
R23
10Ω/1%
C22
100nF/50V
D11
BAS40-04-HE3-08
D12
BAS40-04-HE3-08
470Ω/1%/0805
R17
CondutorPP
R21
10Ω/1%
ADC_CondutorPP
C20
C21
100nF/50V
ADC_VA
C23
C24
D14
BAS40-04-HE3-08
D15
BAS40-04-HE3-08
TC_IFASE
R30
10Ω/1%
C26
100nF/50V
ADC_IFASE
TC_IDIFF
R31
C27
ADC_IDIFF
VtrAC
R26
27kΩ/1%/0805
R27
33kΩ/1%/0805
R32
10K/1%
10k/(10k+60k)=10/70
U13.2
R28
10K/1%
R33
10K/1%
R29
10Ω/1%
C25
100nF/50V
D13
ADC_VGRID
ADC_VGRID=1/2(5V+10/70*VtrAC)
2.5V+5/70*VtrAC
VtrACmax = ~35Vpk
S1
Terminal 6.3mm
SW1
PEC11R-4015F-S0024
CLK
DT
SW
R42
R43
R44
C29
C30
C31
U10
ARDUINO ATMEGA2560
Arduino MEGA
GPIO_PWM_D11
GPIO_RLY
D17_BTPWR
D18_TX1
D19_RX1
ADC_VGRID
ADC_IFASE
ADC_IDIFF
ADC_+14V
ADC_VA
ADC_CondutorPP
+2V5BUFF
+2V5BUFF2
+2V495
RLYcmd
GPIO_RLY
R35
1kΩ/1%/0805
Q1
D882_C9634
27kΩ/1%/0805
R37
R38
10K/1%
LED_RED3
+12VRLY
RLY1
RLY2
RELAY-12V-30A
D16
D17
M1
Fase
Neutro
VDR
R40
C28
100nF/400V
U16
U18
U20
T44008
CondutorPE
TC_IFASE
TC_IDIFF
R34
10kΩ/3296W
R36
470Ω/1%/0805
R39
R41
Dimensionar
de acordo
com o TC
TC1
TC2
2-PIN-1.5MM2
P1
P2
P3
P4
FaseVE
FaseVE1
NeutroVE
NeutroVE1
U14
U15
U17
U19
U21
BLUETOOTH
USART1 => 115200bps
BT MODULE
Header-Female-2.54_1x6
P19
+5Vcomm
SB1
SB2
SB11
BT_EN
R72
R85
R86
R87
680/1%
R88
R89
R90
R91
100K
Q7
Q8
SS8050_C2150
SS8550_C8542
10uF/25V
100nF
100uF/25V
C43
C44
C45
BTEN: DEPENDENDO DO HC
ALTO: ATIVA MODO COMANDO
(AT,38400bps)
BAIXO: ATIVA MODO DADOS
RETIRAR RESISTOR/JUMPER
PARA MODO DADOS.
DESATIVAR A ENERGIA DO
BT PARA FASES DE
CALIBRACAO DO ADC
TITLE:
Sheet_PCBCharger
REV: 1.0
Company: LDSB Tech
Sheet: 1/1
Date: 2026-09-14
Drawn By: daniel_bezerra
EasyEDA

ARDUINO MEGA PINOUT DIAGRAM



TITLE: Sheet_MEGAPinout		REV: 1.0
EasyEDA	Company: LDSB Tech	Sheet: 1/1
	Date: 2026-09-18 Drawn By: daniel_bezerra	